

COURSE OUTLINE

Department/ Faculty:	Software Engineering/Computing	Page:	1 of 8
Course code:	SECJ2013	Academic Session/Semester:	2024-2025/1
Course name:	Data Structures and Algorithms	Pre/co requisite (course name and code, if applicable):	Programming Technique 1
Credit hours:	3		Programming Technique 2

Course synopsis	This course emphasis on data structure concepts theoretically and practically with detail algorithms for each of data structure. Students will learn abstract data type concepts using class and apply the concept in the implementation of data structures. Apart from it, student will learn recursive as a programming style and algorithm efficiency analysis with Big O notation. Various sorting and searching techniques will be discussed as data structure operations. Efficiency analysis of each algorithm will also be explained. Further, students will be exposed to linear data structures such as linked lists, stack and queue, while tree and binary search tree will be introduced as non-linear data structures. Along the course, students should be able to implement and apply the theory and concepts of data structure in the assignments and mini project which are conducted in group.			
Course coordinator	Dr. Mohd Razak bin Samingan			
Course lecturer(s)	Name	Section	Tel	E-mail (@utm.my)
	Dr. Nur Eiliyah @ Wong Yee Leng	01		nureiliyah
	Dr. Zuraini binti Ali Shah	02		aszuraini
	Dr. Mohd Razak bin Samingan	03		mrarak
	Dr. Nur Eiliyah @ Wong Yee Leng	04		nureiliyah
	Mr. Norizam bin Katmon	05		rizam
	Dr. Ruhaidah binti Samsudin	06		ruhaidah
	Dr. Mohd Razak bin Samingan	07		mrarak
	Dr. Zuraini binti Ali Shah	08		aszuraini
	Dr. Pang Yee Yong	09		yeeyongpang
	Dr. Fazliaty Edora binti Fadzli	10		fazliaty.edora
	Dr. Fazliaty Edora binti Fadzli	11		fazliaty.edora

Mapping of the Course Learning Outcomes to the Programme Learning Outcomes, Teaching & Learning (T&L) methods and Assessment methods:

No.	Course Learning Outcome	*Program Learning Outcome	Weight (%)	**Taxonomies and generic skills	T&L methods	***Assessment methods
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Prepared by:

Name:	Nor Bahiah Hj Ahmad (Course Owner)
Signature:	
Date:	24 August 2017

Certified by:

Name:	PM. Dr. Siti Zaiton Mohd Hashim (Head of Department)
Signature:	
Date:	

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CLO1	Students are able to choose the most suitable data structures and analyse the most effective algorithm for solving certain problems.	PLO1 (KW)	35	C3	Lecture, active learning	Q, T, F
CLO2	Apply effectively data structure concepts in programming.	PLO2 (AP)	45	C3	Lab work	F, L, LT
CLO3	Develop medium scale programs as a group mini project by applying data structure concepts in problem solving.	PLO2 (AP) PLO7(TW)	15 5	C3, TW2	Project-based learning	A, GR, PR
*Program Learning Outcome PLO1 - Ability to acquire and apply theory and principles of computer science and equip with social science and personal development knowledge. PLO2 - Ability to design and construct computer programs using standard approaches. PLO7 - Ability to work effectively in a team						

****Taxonomies and Generic Skills**

C2 - Comprehension (describe, discuss, explain, locate, paraphrase, give example, translate)

C3 - Application (apply, carry out, demonstrate, illustrate, prepare, solve, use)

TS2 - Taking roles as Leader and Member

***T – Test; Q – Quiz; LT-Lab Test; L – Lab, A – Assignment, GR – Group Mini Project; PR – Presentation; F – Final Exam etc.

Details on Innovative T&L practices:

No.	Type	Implementation
1.	Active learning	Conducted through in-class activities
2.	Independent Study	E-learning, MOOCs
3.	Project-based learning	Conducted through case study assignment and mini projects. Tasks are given in sequential steps throughout the semester. Students in a group of 3 are given a case study that requires them to use data structure concepts to solve certain problems efficiently.

Weekly Schedule:

Week 1 06 - 12 Oct	Chapter 1: Introduction to Data Structures and Algorithms ● Introduction to Data Structure ● Types of Data Structure ● Algorithms ● Programming Principle
Week 2 13 - 19 Oct	Chapter 2: Abstract Data Type, C++ Revision and Advanced File Operations (Lab 1) ● Abstract Data Type Concepts ● C++ Revision ● C++ class ● Pass objects to functions, class as return value for functions

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	• Array of class and pointer to class • Advanced File Operations • File Operations • Passing File Stream Objects to Functions • Member Functions for Reading and Writing Files • Binary Files • Random-Access Files
Week 3 20 – 26 Oct	Chapter 3 & 4: Recursive and Algorithm Efficiency (Quiz 1) • Recursive Principle and design • Tracing Recursive Algorithms • Algorithm Efficiency • Analysis order of magnitude, Big O notation
Week 4 27 Oct – 2 Nov <i>Deepavali</i> <i>(31 Oct – Thursday)</i> Week 5 3 – 9 Nov	Chapter 5: Sorting (Lab 2, Assignment 1) • Simple Sort • Bubble Sort • Selection Sort • Insertion Sort • Sorting Analysis • Advanced Sort (Quiz 2) • Merge Sort • Quick Sort
Week 6 10 – 16 Nov	Chapter 6: Searching • Sequential Search • Binary Search
Week 7 17 – 23 Nov	Chapter 7: Linked List (Lab 3, Assignment 2) • Pointer Concepts • Introduction to Linked lists • Linked lists operations • Types of Linked List Test 1 (MCQ & SHORT ANSWER) TOPIC: (Chapter 1 - Chapter 5) Date: Wednesday, 20 Nov 2024 Time: 8.00 pm – 10.00 pm Location: BK1 - BK7
Week 8 24 – 30 Nov	MID TERM BREAK
Week 9 01 – 07 Dec	Linked List (continued) • Linked List Implementations ○ Declaring Nodes and Linked Lists class ○ Insert Node, Delete Node, Find Node, Print Node

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Week 10 08 – 14 Dec Week 11 15 – 21 Dec	Chapter 8: Stack • Introduction to Stack • Stack Operations ◦ push(), pop() • Stack implementations ◦ Stack Array ◦ Stack Linked List • Stack Applications ◦ Infix, Postfix and Prefix
Week 12 22 – 28 Dec <i>Christmas</i> <i>(25 Dec -</i> <i>Wednesday)</i>	Chapter 9: Queue Lab 4, Mini Project, Quiz 3 • Introduction to Queue • Queue Implementations ◦ Array Implementation (Linear and Circular) ◦ Linked List Implementation (Linear and Circular Queue)
Week 13 29 Dec – 04 Jan & Week 14 05 – 11 Jan	Chapter 10: Tree Quiz 4, Lab Test • Introduction to Tree • Terms related to Tree concepts • Binary Search Tree Binary Search Tree Implementations • Declaring Tree node, Tree class • Create Node, Insert Node, Delete Node, Search Node • Tree Traversals TEST 2 (LAB TEST) TOPIC: Linked List and Stack Date: Wednesday, 08 Jan 2025 Time: 8.00 pm - 10.00 pm Location: MPK1- MPK10, CGMTL, CCNA, CVTL, MCASE
Week 15 12 – 18 Jan	Final Exam Revision (Searching, Linked List, Stack, Queue, Tree) Group Project Demo/Presentations
Week 16 19 – 25 Jan	REVISION WEEK
26 Jan – 15 Feb	EXAM WEEKS
16 Feb – 15 Mac	FINAL BREAK FOR SEM 1

Transferable skills (generic skills learned in course of study which can be useful and utilised in other settings):

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Team working and Presentation

Student learning time (SLT) details:

Distribution of student Learning Time (SLT) Course content outline			Teaching and Learning Activities		TOTAL SLT
	Guided Learning (Face to Face)		Guided Learning Non-Face to Face	Independent Learning Non-Face to face	
CLO	Lect	Tutorial/ Lab			
CLO 1-3	28h	28h	10	36	102h

Continuous Assessment		PLO	Percentage	Total SLT
1	Lab (4)	AP	10	Inclusive in CLO1 (4h)
2	Test 1	KW	15	2h
3	Lab Test	AP	15	2h
4	Quiz (2)	KW	10	1h
5	Mini Project	AP / TW	10	Inclusive in CLO2, CLO3 (4h)
6	Assignment (2)	AP / TW	10	Inclusive in CLO2, CLO3 (16h30m)
Final Assessment			Percentage	Total SLT
1	Final Examination	KW/ AP	30	3h
Grand Total SLT				120h

Special requirement to deliver the course (e.g: software, nursery, computer lab, simulation room):

Software: C++ Compiler

Learning resources:

Main references

Nor Bahiah, Dayang Norhayati and Zalmiyah. Data Structures and Algorithms, Third Edition. Universiti Teknologi Malaysia, 2014. (Compulsory)

Additional references

Frank M Carano and Timothy Henry, "Data Abstraction and Problem Solving with C++", Walls and Mirrors, Seventh Edition, (2017), Pearson. **Online**

Course Website:

Utm E-learning at <http://elearning.utm.my>

Massive Open Online Learning Data Structures and Algorithms:

<https://www.openlearning.com/courses/SCSJ2013x>

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Academic honesty and plagiarism:

Copying of work (texts, lab results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of **zero** for the assignment and exams and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (Course policy, any specific instruction etc.):

1. Attendance is compulsory and will be taken in every lecture session. Student with less than 80% of total attendance is not allowed to sit for final exam.
2. Students are required to behave and follow the University's dressing regulation and etiquette all the time.
3. Exercises and tutorial will be given in class and some may be taken for assessment. Students who do not do the exercise will lose the coursework marks for the exercise.
4. Assignments must be submitted on the due dates. Some points will be deducted for late submissions. Assignments submitted three days after the due date will not be accepted.
5. Make up exam will not be given, except to students who are sick and submit medical certificate confirmed by UTM panel doctors. Make up exam can only be given within one week of the initial date of exam.

No	Assessment	PLO1	PLO2		PLO7	TOTAL
		CLO1	CLO2	CLO3	CLO3	
1	LAB (4)		10			10
2	TEST 1	15				15
3	Lab TEST		15			15
4	FINAL EXAM	10	20			30
5	Assignment (2)			8	2	10
6	Quiz (2)	10				10
7	Project			7	3	10
TOTAL PLO		35	45	15	5	100

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